

Quantum Information Theory

Lesson 7: Quantum Leftover Hashing Lemma

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Remember: Classical Compression

The key insight when we discussed classical compression was to look at any input message $\mathbf{w}$ as a stream of i.i.d. samples of a random variable X called a memoryless source. This X produced strings of message tokens $m_0, \dots, m_n \in \mathcal{M}$, so that $\mathbf{w}$ would be one of the possible message strings obtained from X .

To see how much that information could be compressed, we considered a variable-length encoder $f : \mathcal{M} \rightarrow \mathbb{Z}_d^*$ such as the Huffman algorithm, and we wanted to shrink the expected codeword length while still being able to invert (decode) f .

Shannon's source coding theorem looked at what happened when we tried to stream possibly infinitely long streams of samples of X . It stated that the optimal average codeword length satisfies

$$\lim_{n \rightarrow \infty} \frac{1}{n} |\mathbb{E}[L] - nH(X)| = 0,$$

a good asymptotic approximation of the true optimum, that for finite n is numerically computed analyzing the Huffman algorithm.

Remember: Classical Compression

To prove Shannon's coding theorem, we looked at what happens if we try to use less dits per message token, and we found that if we used less then there is no decoder that we can use to invert f .

To handle the happy path, we noticed that in a string of n tokens, the frequency of each possible token n_i is clustered around $np_i \pm \epsilon$ so to give our encoder some leeway we were even open to let our decoder fail when the input was not as just described, ϵ -typical. If we let our decoder fail with probability π then we could ignore ϵ -atypical inputs with $\epsilon = 1/\sqrt{\pi n}$. But, alas, this was not enough.

Indeed, as long as $\pi \neq 1$, we would need $\mathbb{E}[L] \geq nH(X) + o(n)$. More precisely, we found that if we are open to let our encoder use on average $H(X) + \delta$ dits per token, then there is an $\epsilon \leq \delta/C_{X,d}$ and a minimal packed amount $n \geq 1/(\pi\epsilon^2)$ of letters for which such an encoder can be found to fail with probability less than π .

Entropy - how uncertain are you?

Let's remember...

You follow the weather in two different cities, one is set in a desert, the other is in a temperate climate.

- ▶ Desert city: the weather is sunny the whole year. There is no new information when learning how the weather is going to be tomorrow, as it is also going to be sunny. This means there is *no uncertainty* regarding the weather. $\rightarrow H(X) = 0$
- ▶ Temperate city: the weather is unpredictable. It can be sunny, rainy, foggy, cloudy, or stormy. If we consider the weather to be a random variable with uniform distribution, then we can say that we are *entirely uncertain* regarding the next day's weather. $\rightarrow H(X) = \log n$

This notion was introduced by Claude Shannon in 1948 - everybody knows Shannon entropy.

Plot twist: There are multiple types of entropy, each measuring uncertainty from a different angle. *Not all entropies are equal — and choosing the right one is half the battle.*

Rényi entropy - $\alpha = 2$

Alfréd Rényi wanted to generalize the notion of entropy in such a way to preserve the additivity of independent events. This is known as **Rényi entropy**, and for different α we can derive different known entropies such as **collision entropy**, for $\alpha = 2$.

$$H_{\alpha}(X) = \frac{1}{1 - \alpha} \log \left(\sum_{i=1}^n p_i^{\alpha} \right),$$

which becomes

$$H_2(X) = -\log \left(\sum_{i=1}^n p_i^2 \right).$$

What could the sum obtained represent?

Collision Entropy

The sum $\sum_{i=1}^n p_i^2$ represents the collision probability - the probability that two independent samples from the same distribution produce the same value.

$$P_{collision} = \sum_{i=1}^n p_i^2 = Pr[X = X'],$$

where X and X' represent independent copies of the same random variable. We now ask the question: *If I sample from the same distribution twice independently, how likely am I to get the same outcome?*

- ▶ If the distribution is spread out, which means it is hard to hit the same value twice (i.e., lots of randomness), that results in **low collision probability**, or **high collision entropy**.
- ▶ If the distribution is concentrated on a few values, which means it is more likely to hit the same value twice (i.e., little randomness), that results in **high collision probability**, or **low collision entropy**.

Min-Entropy and Smooth-Min Entropy

Let's now consider what happens when $\alpha \rightarrow \infty$:

$$H_{\infty}(X) = \min_i (-\log p_i) = -(\max_i \log p_i) = -\log \max_i p_i,$$

but intuitively, what does it mean to use the highest probability of the distribution?

Let's assume we have an adversary Eve, who knows the probability distribution of the random variable X . What is then the probability of Eve guessing the value of X in a single try?

The optimal strategy is for Eve to guess the most likely event (i.e., the one with the highest probability). We can think about

Min-Entropy as the most pessimistic measure of entropy.

- ▶ High guessing probability $\rightarrow$ low min-entropy
- ▶ Low guessing probability $\rightarrow$ high min-entropy

Conditional Min-Entropy

Consider a joint probability distribution $P(X, Y)$ where X and Y can each take values in $\{0, 1\}$. The joint probabilities are given by:

- ▶ $P(X = 0, Y = 0) = 0.2$
- ▶ $P(X = 0, Y = 1) = 0.3$
- ▶ $P(X = 1, Y = 0) = 0.1$
- ▶ $P(X = 1, Y = 1) = 0.4$

Compute the conditional min-entropy $H_{\min}(X|Y)$.

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

$$H_{\min}(X|Y) = -\log \sum_y P(Y = y) \max_x P(X = x|Y = y)$$

Quantum side information

Quantum side information refers to any type of quantum mechanical data or correlations that an observer has, which can be used in processing, compressing, or securing information in another system.

Consider Z a classical random variable with the probability distribution p_z , and E a quantum system whose state conditioned on $Z = z$ is given by the density operator $\rho_E^{[z]}$.

$$\rho_{ZE} = \sum_{z \in Z} p_z |z\rangle \langle z|_Z \otimes \rho_E^{[z]}.$$

Z is *uniform conditioned* on E if

$$\rho_{ZE} = \omega_Z \otimes \rho_E, \text{ where } \omega_Z = \mathbb{I}_Z / |Z|.$$

Z is Δ -close to *uniform conditioned* on E if there exists a state σ_E on E such that:

$$\frac{1}{2} \|\rho_{ZE} - \omega_Z \otimes \sigma_E\| \leq \Delta.$$

Back to Min-Entropy

Consider the following notations:

- ▶ $\mathcal{H}$: finite-dimensional Hilbert space;
- ▶ $\mathcal{P}(\mathcal{H})$: set of positive semi-definite operators on $\mathcal{H}$;
- ▶ $S_=(\mathcal{H}) = \{\rho \in \mathcal{P}(\mathcal{H}) : \text{Tr}(\rho) = 1\}$: set of normalized quantum states
- ▶ $S_{\leq}(\mathcal{H}) = \{\rho \in \mathcal{P}(\mathcal{H}) : 0 < \text{Tr}(\rho) \leq 1\}$: set of sub-normalized quantum states

The generalized definition of the min-entropy of A conditioned on B where $\rho_{AB} \in S_{\leq}(\mathcal{H}_{AB})$ is:

$$H_{\min}(A|B)_{\rho} := \max_{\sigma_B \in S_=(\mathcal{H}_B)} \sup \left\{ \lambda \in \mathbb{R} : \rho_{AB} \leq 2^{-\lambda} \mathbb{I}_A \otimes \sigma_B \right\}.$$

Side quest: Purified distance

Consider two quantum states ρ and σ . For these two quantum states, we can define the *trace distance* as:

$$D(\rho, \sigma) = \frac{1}{2} \text{Tr} |\rho - \sigma|, \text{ where } |A| = \sqrt{A^\dagger A}$$

The trace distance is a metric as it possesses some important qualities, namely:

- ▶ $D(\rho, \sigma) = 0 \Leftrightarrow \rho = \sigma$
- ▶ $D(\rho, \sigma) = D(\sigma, \rho)$
- ▶ $D(\rho, \tau) \leq D(\rho, \sigma) + D(\sigma, \tau)$

At the same time, we can define the generalized fidelity of the two quantum states as:

$$F(\rho, \sigma) = \text{Tr} |\sqrt{\rho}\sqrt{\sigma}| + \sqrt{(1 - \text{Tr}(\rho))(1 - \text{Tr}(\sigma))}.$$

Thus, we can define the purified distance as:

$$P(\rho, \sigma) = \sqrt{1 - F(\rho, \sigma)^2}$$

Back to Smooth-min Entropy

Let's define the ϵ -ball of states close to $\rho \in S_{\leq}(\mathcal{H})$ as

$$\mathcal{B}^{\epsilon}(\rho) = \{\tilde{\rho} \in S_{\leq}(\mathcal{H}) : P(\rho, \tilde{\rho}) \leq \epsilon\}.$$

In this case, the smooth-min entropy of A conditioned on B is defined as:

$$H_{\min}^{\epsilon}(A|B)_{\rho} = \max_{\tilde{\rho}_{AB} \in \mathcal{B}^{\epsilon}(\rho_{AB})} H_{\min}(A|B)_{\tilde{\rho}}$$

Two-Universal Hash Functions

Hash function: any function that can be used to map data of arbitrary length to fixed-size values.

Let $p = 5$ (a prime number), and the working space be $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$. We want to hash two-element inputs to one-element outputs, so we define:

$$f_a : \mathbb{Z}_5^2 \rightarrow \mathbb{Z}_5, f_a(x_1, x_2) = a_1x_1 + a_2x_2 \mod 5$$

where $a = (a_1, a_2) \in \mathbb{Z}_5^2$ is the seed. There are exactly 25 functions that satisfy these conditions, thereby forming a *family of hash functions*. Take $a = (2, 3)$ as an example: $f_{(2,3)}(x_1, x_2) = 2x_1 + 3x_2 \mod 5$.

At the same time, let's consider inputs $(1, 1)$ and $(4, 4)$:

$f_{(2,3)}(1, 1) = 0$ and $f_{(2,3)}(4, 4) = 0$, so $f_{(2,3)}(1, 1) = f_{(2,3)}(4, 4)$.

This is called a **collision**.

A family $\mathcal{F}$ of functions from $\mathcal{X}$ to $\mathcal{Z}$ is considered to be two-universal if, for any pair of distinct inputs x and x' , and for any $f \in \mathcal{F}$ chosen randomly, the probability of a collision $f(x) = f(x')$ is not larger than $\frac{1}{|\mathcal{Z}|}$.

In the above example, we can demonstrate that the family we considered is two-universal. For any pair of inputs and for any seed, we can show that

$$P_{\text{collision}}[f_a(x_1, x_2) = f_a(x'_1, x'_2)] = \frac{5}{25} = \frac{1}{5} = \frac{1}{|\mathcal{Z}_5|}$$

As an extension, a family of functions $\mathcal{F}$ from $\mathcal{X}$ to $\mathcal{Z}$ is called δ -almost two-universal if:

$$P_{\text{collision}}[f_a(x) = f_a(x')] \leq \delta,$$

for any $x \neq x'$.

Randomness extraction

Consider a random variable X over $\{0, 1\}^n$ with min-entropy $H_{\min}(X) = k$, and let U_d be a uniformly random seed independent of X . A function

$$\text{Ext} : \{0, 1\}^n \times \{0, 1\}^d \rightarrow \{0, 1\}^m$$

is a (k, ε) -extractor if

$$\frac{1}{2} \text{Tr} |\text{Ext}(X, U_d) - U_m| \leq \varepsilon.$$

Now it all comes together in...

Classical Leftover Hashing Lemma

Let X and E be random variables and $\mathcal{F}$ be a two-universal family of hash functions with domain $\mathcal{X}$ and range $\{0, 1\}^l$. Then, on average over the choices of f from $\mathcal{F}$, the distribution of the output $Z = f(X)$ is Δ -close to uniform conditioned on E , where

$$\Delta = \frac{1}{2} \sqrt{2^{l-H_{\min}(X|E)}}.$$

Quantum Leftover Hashing Lemma

Statement: Let X be a random variable, let E be a quantum system, and let $\mathcal{F}$ be a δ -almost two-universal family of hash functions from $\mathcal{X}$ to $\{0, 1\}^l$. Then, on average over the choices of $f \in \mathcal{F}$, the output $Z = f(X)$ is Δ -close to universal conditioned on E where

$$\Delta = \inf_{\epsilon > 0} \frac{1}{2} \sqrt{(2^l \delta - 1) + 2^{l - H_{\min}(X|E) + \log(\frac{2}{\epsilon^2} + 1)}} + \epsilon.$$

Furthermore, if $\delta \leq 2^{-l}$, meaning if $\mathcal{F}$ is two-universal, then

$$\Delta = \frac{1}{2} \sqrt{2^{l - H_{\min}(X|E)}}.$$

Privacy amplification

Imagine that Alice and Bob use a QKD protocol (such as BB84). They exchange quantum states over a channel to establish a shared secret key. QKD promises that any eavesdropping by Eve disturbs the quantum states and can be detected. After the protocol, Alice and Bob share a raw key. The raw key is a string of bits that can be either partially secret (Eve might have intercepted some information), or partially correlated (resulting either from errors in the channel or from Eve's interference, yet still leading to Alice and Bob disagreeing on the key). Let's consider a joint state ρ_{KE} , where K is Alice and Bob's key and E is Eve's quantum system. If $H_{\min}(K|E)_{\rho} \approx 0$, Eve can guess K with high probability thanks to her E .

If $\rho_{KE} \approx \omega_Z \otimes \rho_E$, then the key is uniformly random and independent of Eve's system, thus hard (if not impossible) to guess. Privacy amplification is the fix: we can extract from our raw key a compressed key with enough genuine randomness to make it secure, which is guaranteed by the Quantum Leftover Hashing Lemma.

Quantum State Discrimination

Before creating a procedure to compress quantum states, we would first need to be able to tell if what we decoded was any good.

For a toy setting, let's say what we decoded is, with probability p , the ρ that we encoded, and, with probability $1 - p$, some fixed σ .

To distinguish them, we use a POVM with operators $\mathbf{F}$ and $\mathbf{I} - \mathbf{F}$. The probability to correctly identify what was sent is

$$\begin{aligned} p \operatorname{Tr}(\mathbf{F}\rho) + (1 - p) \operatorname{Tr}((\mathbf{I} - \mathbf{F})\sigma) &= 1 - p + \operatorname{Tr}(\mathbf{F}(p\rho - (1 - p)\sigma)) \\ &= p - \operatorname{Tr}((\mathbf{I} - \mathbf{F})(p\rho - (1 - p)\sigma)) \end{aligned}$$

and, if we note that the eigenvalues of $\mathbf{F}$ must be ≤ 1 , this quantity is maximized by taking $\mathbf{F}$ to be the projector on the positive eigenspace of $p\rho - (1 - p)\sigma$, known as the Helstrom Measurement. Taking $p\rho - (1 - p)\sigma = \sum_i \lambda_i |\psi_i\rangle \langle \psi_i|$ the two expressions become

$$1 - p + \sum_{\lambda_i > 0} \lambda_i = p - \sum_{\lambda_i < 0} \lambda_i = \frac{1}{2} + \frac{1}{2} \sum_i |\lambda_i| = \frac{1}{2} + \frac{1}{2} \|p\rho - (1 - p)\sigma\|_{\operatorname{Tr}},$$

known as the Helstrom bound.

Quantum Pinsker Inequality

Notice that the lower the Helstrom bound is, the more similar ρ and σ are. In particular, if we take $p = 1 - p = 1/2$, the Helstrom bound becomes $\frac{1}{2} + \frac{1}{4} \|\rho - \sigma\|_{Tr}$, which we can relate to $D(\rho\|\sigma)$.

So, let's consider all state ρ, σ such that $\|\rho - \sigma\|_{Tr}$ and consider what their minimal Umegaki divergence might be.

To start, consider the channel that performs the Helstrom measurement and records its value in a qubit state.

Letting $\rho - \sigma = \sum_i \lambda_i |\psi_i\rangle \langle \psi_i|$ we thus consider

$$\mathcal{E}(\mathbf{M}) = \sum_{\lambda_i > 0} |0\rangle \langle \psi_i| \mathbf{M} |\psi_i\rangle \langle 0| + \sum_{\lambda_i < 0} |1\rangle \langle \psi_i| \mathbf{M} |\psi_i\rangle \langle 1|,$$

and since $\mathcal{E}$ is a channel it lowers the Umegaki divergence.

However, as it turns out the Helstrom bound for discriminating $\mathcal{E}(\rho)$ and $\mathcal{E}(\sigma)$ is $\|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_{Tr} = \|\mathcal{E}(\rho - \sigma)\|_{Tr} = \|\rho - \sigma\|_{Tr}$. It follows that the minimal Umegaki divergence for a given trace distance is attained for diagonal qubit states.

Quantum Pinsker Inequality

For diagonal states ρ, σ the trace distance only depends on their (probabilistic) diagonal vectors $\mathbf{p}, \mathbf{q}$, $\|\rho - \sigma\|_{Tr} = \|\mathbf{p} - \mathbf{q}\|_1$, so we can borrow a result from classical information theory known as the Pinsker inequality which states that

$$\|\mathbf{p} - \mathbf{q}\|_1 \leq \sqrt{2 \ln 2 D_{KL}(\mathbf{p} \parallel \mathbf{q})},$$

and so we find that for any quantum states ρ, σ

$$\begin{aligned} \|\rho - \sigma\|_{Tr} &= \|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_{Tr} = \|\mathbf{p} - \mathbf{q}\|_1 \\ &\leq \sqrt{2 \ln 2 D_{KL}(\mathbf{p} \parallel \mathbf{q})} = \sqrt{2 \ln 2 D(\mathcal{E}(\rho) \parallel \mathcal{E}(\sigma))} \\ &\leq \sqrt{2 \ln 2 D(\rho \parallel \sigma)}, \end{aligned}$$

a relation known as the quantum Pinsker inequality.

We thus successfully related another practically relevant quantity, the maximal probability to distinguish between two states, to an entropy, here the quantum relative entropy or Umegaki divergence.

Entanglement-Assisted Quantum State Discrimination

Mixed states are often a result of tracing out part of an entangled state. Let's say Alice's system has a density matrix either ρ or σ with equal probability, but her system is entangled with Bob's, who is thus able to help Alice. Let's say they share $|\phi_\rho\rangle$ or $|\psi_\sigma\rangle$.

Since there are always $\theta, |\xi\rangle$ such that $|\phi_\rho\rangle = \cos \theta |\psi_\sigma\rangle + \sin \theta |\xi\rangle$,

$$|\phi_\rho\rangle \langle \phi_\rho| - |\psi_\sigma\rangle \langle \psi_\sigma| \sim \begin{bmatrix} \cos^2 \theta - 1 & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$$

with eigenvalues $\pm \sqrt{|\sin^2 \theta (\cos^2 \theta - 1) - \cos^2 \theta \sin^2 \theta|} = \pm \sin \theta$ so the Helstrom bound is $\frac{1}{2}(1 + |\sin \theta|) = \frac{1}{2}(1 + \sqrt{1 - |\langle \phi_\rho | \psi_\sigma \rangle|^2})$.

So, the maximal probability that Alice can discern what was sent depends on the minimal overlap between two purifications, and conversely, the minimal probability that Alice can discern what was sent depends on the maximal overlap between two purifications, a quantity known as the fidelity of the two quantum states.

Uhlmann's Theorem

As it turns out, for any ρ, σ we can compute the fidelity $F(\rho, \sigma)$ as

$$\max \left\{ |\langle \phi_\rho | \psi_\sigma \rangle|^2 \mid \text{Tr}_B |\phi_\rho\rangle \langle \phi_\rho| = \rho, \text{Tr}_B |\psi_\sigma\rangle \langle \psi_\sigma| = \sigma \right\} = \text{Tr}^2 |\sqrt{\rho} \sqrt{\sigma}|.$$

To see why, let's fix some decompositions. Let's take

$$\rho = \sum_i p_i |a_i\rangle \langle a_i|; \quad \sigma = \sum_j q_j |b_j\rangle \langle b_j|;$$

$$|\phi_\rho\rangle = \sum_i \sqrt{p_i} |a_i\rangle \otimes |x_i\rangle; \quad |\psi_\sigma\rangle = \sum_j \sqrt{q_j} |b_j\rangle \otimes |y_j\rangle$$

$$\begin{aligned} \Rightarrow |\langle \phi_\rho | \psi_\sigma \rangle| &= \left| \sum_{i,j} \sqrt{p_i q_j} \langle a_i | b_j \rangle \langle x_i | y_j \rangle \right| \\ &= \left| \sum_{i,j,k,l,m} \langle a_k | \underbrace{\sqrt{p_i} |a_i\rangle \langle a_i|}_{\sqrt{\rho}} \underbrace{\sqrt{q_j} |b_j\rangle \langle b_j|}_{\sqrt{\sigma}} \underbrace{\langle b_l | \langle y_l | \overline{x_m} \rangle \langle a_m |}_{\mathbf{U}} |a_k\rangle \right| \\ &= |\text{Tr}(\sqrt{\rho} \sqrt{\sigma} \mathbf{U})| \leq \text{Tr}(\sqrt{(\sqrt{\rho} \sqrt{\sigma})(\sqrt{\rho} \sqrt{\sigma})^\dagger}) = \text{Tr} |\sqrt{\rho} \sqrt{\sigma}|, \end{aligned}$$

where we used the triangle inequality $|\text{Tr}(\mathbf{AU})| \leq \text{Tr} |\mathbf{A}|$ is saturated when $\mathbf{AU}$ is positive semi-definite, e.g. when $|x_i\rangle = |y_i\rangle$.

Entanglement Fidelity

Let's now say that we want to develop a quantum compression algorithm that does not degrade quality of service.

Essentially, we would want Alice to be unable to tell that her states were compressed and decompressed even if she tried.

Alice can get help from any party whom she holds entanglement with, and so what we intuitively want to maximize is the quantum fidelity between ρ and $\mathcal{E}(\rho)$, that is $F(\rho, \mathcal{E}(\rho)) = \text{Tr}^2 \left| \sqrt{\rho} \sqrt{\mathcal{E}(\rho)} \right|$. The problem is that the usual fidelity $F(\rho, \sigma)$ maximizes over both purifications of ρ, σ , whereas our compression $\mathcal{E}$ only acts on Alice's system so Bob does not have free choice over the state of his subsystem, hence we need a more refined type of fidelity.

So, instead, we are going to consider the entanglement fidelity,

$$\begin{aligned} F(\rho, \mathcal{E}) &= F(\mathbf{I}_{Bob} \otimes \mathcal{E}_{Alice}(|\psi_\rho\rangle \langle \psi_\rho|), |\psi_\rho\rangle \langle \psi_\rho|) \\ &= \langle \psi_\rho | \mathbf{I}_{Bob} \otimes \mathcal{E}_{Alice}(|\psi_\rho\rangle \langle \psi_\rho|) | \psi_\rho \rangle, \end{aligned}$$

where any chosen purification $|\psi_\rho\rangle$ produces the same result.

Quantum Fano Inequality

When, as before, σ is pure i.e. $\sigma = |\phi\rangle\langle\phi|$ we find

$$\begin{aligned} F(\rho, \sigma) &= \text{Tr} |\sqrt{\rho}\sqrt{\sigma}| = \text{Tr} \sqrt{(\sqrt{\rho}\sqrt{\sigma})^\dagger (\sqrt{\rho}\sqrt{\sigma})} = \text{Tr} \sqrt{\sigma}^\dagger \rho \sqrt{\sigma} \\ &= \text{Tr} |\phi\rangle\langle\phi| \rho |\phi\rangle\langle\phi| = \langle\phi| \rho |\phi\rangle, \end{aligned}$$

numerically equal to the probability that ρ would collapse into $|\phi\rangle$. If Alice and Bob try to evaluate the entanglement fidelity $F(\rho, \mathcal{E})$, they thus can simply measure upon $\rho' = \mathbf{I} \otimes \mathcal{E}(|\psi_\rho\rangle\langle\psi_\rho|)$ a POVM with operators $\{|\psi_\rho\rangle\langle\psi_\rho|, \mathbf{I} - |\psi_\rho\rangle\langle\psi_\rho|\}$ and they may for example record the outcome in a single-qubit (diagonal) state ρ'' .

Looking at the entropy exchange when ρ goes through $\mathcal{E}$ we get

$$\begin{aligned} H(\rho, \mathcal{E}) &= H(\rho') - H(|\psi_\rho\rangle\langle\psi_\rho|) \\ &= \log d^2 - D(\rho' \mid \mathbf{I}_d/d^2) - 0 \\ &\leq \log d^2 - D(\rho'' \mid \frac{1}{d^2} |0\rangle\langle 0| + \frac{d^2 - 1}{d^2} |1\rangle\langle 1|) \\ &= h(F(\rho, \mathcal{E})) + (1 - F(\rho, \mathcal{E})) \log(d^2 - 1), \end{aligned}$$

known as the Quantum Fano Inequality.

Schumacher's Source Coding Theorem

With our new tricks, we can now look at quantum compression. As with Shannon's source coding theorem, what we are going to be compressing is not a single state but a (possibly infinite) stream of i.i.d. samples of a random variable with matrix $\rho \in \mathcal{B}(\mathcal{H})$.

But, to define encoder and decoder channels we need to know how many such samples n we are processing at a time, and how many qubits m we may output. We say that a pair of channels

$$\mathcal{E} : \mathcal{B}(\mathcal{H}^{\otimes n}) \rightarrow \mathcal{B}(\mathbb{C}^{2^{\otimes m}}); \mathcal{D} : \mathcal{B}(\mathbb{C}^{2^{\otimes m}}) \rightarrow \mathcal{B}(\mathcal{H}^{\otimes n})$$

is a (n, m, π) compression scheme for ρ if $F(\rho^{\otimes n}, \mathcal{D} \circ \mathcal{E}) \geq 1 - \pi$.

Let's now say we are given samples in bulk ($n \rightarrow \infty$), and we want to know how many qubits R we need per sample on average, i.e. the minimal R such that as $n \rightarrow \infty$ we can always find a $(n, nR + o(n), \pi_n)$ compression scheme with $\pi_n \rightarrow 0$.

Schumacher's Source Coding Theorem states that $\min R = H(\rho)$. Note that lower R would let us violate the Holevo bound.

Schumacher's Source Coding Theorem

As with Shannon's source coding theorem, to show that we can achieve $R = H(\rho)$ we want to look at some ϵ -typical set.

If $\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$, then

$$\rho^{\otimes n} = \sum_{i_1, \dots, i_n} p_{i_1} \dots p_{i_n} |\psi_{i_1}\rangle \langle \psi_{i_1}| \otimes \dots \otimes |\psi_{i_n}\rangle \langle \psi_{i_n}|$$

and so we denote by $\mathcal{I}$ the set of ϵ -typical indices $(i_1, \dots, i_n)$.

Consider now the projector $\mathbf{P} = \sum_{\mathbf{i} \in \mathcal{I}} |\psi_{i_1}\rangle \langle \psi_{i_1}| \otimes \dots \otimes |\psi_{i_n}\rangle \langle \psi_{i_n}|$, and notice that if we measure $\rho^{\otimes n}$ in its eigenbasis, the probability to obtain a result in $\mathcal{I}$ is $\text{Tr}(\mathbf{P}\rho) \geq 1 - \pi$.

Yet, from the proof of Shannon's theorem we know that if we were to simply send such an index it would take $n(H(\rho) + C\epsilon)$ bits, so to have them as base states we would need $n(H(\rho) + C\epsilon)$ qubits. Thus, we choose to encode our state using an isometry that maps the ϵ -typical eigenvectors to base states of our target system.

As it turns out, this is enough to achieve $F(\rho^n, \mathcal{D} \circ \mathcal{E}) \geq 1 - \pi$.

Follow-up Reading

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